Decision Analysis Theory

Decision Making with Multiple Objectives 1

Background Information

- Emphasize shift seen in the 1980s from the dominance of single-objective modeling and optimization toward multiple objectives.
  - Single-objective models developed in 1960s and 1970s are considered by many to be unrealistic, too restrictive, and often inadequate for real-world problems.
- Multiple-Criteria Decision Making (MCDM).
  - Multiple-attribute decision making.
  - Mathematical optimum for a model does not necessarily correspond to the optimum for the real-world problem.
  - Objectives: minimizing cost, maximizing benefits (not necessarily in monetary value), and minimizing risks of various kinds.

Examples

- Buying a car:
  - Size of car
  - Fuel economy of car (miles per gallon)
  - Style of car
  - Price of car
- Graduating student:
  - Starting salary job
  - Location of job
  - Degree of interest toward the work involved
  - Long-term opportunities

Multi-Criteria Decision Making

- Cost and profit are not the only measures of desirability for alternatives selection.
- Other objectives, called intangibles or non-monetary criteria could be used:
  - Customer service, reputation, job satisfaction, safety, employment stability.
- Once the objective is defined, an attribute (criterion) that characterizes the objective must also be defined.
  - Ex: "safety" objective in a vehicle purchase.
    - Criteria: vehicle type, airbags, ABS, etc.

Multi-Criteria Decision Making

- Goal of the decision maker, with respect to multiple objectives and criteria, is to have a single measure of value associated with each alternative.
- Example: select a CAD workstation.
  - Four alternatives: 3 vendors + do nothing choice
  - Seven attributes used in the selection process.

Attribute (criterion)	Alternative			
	Vendor A	Vendor B	Vendor C	Do nothing
Cost of purchasing the system	$115,000	$338,950	$32,000	$0
Reduction in design time	60%	67%	50%	0
Flexibility	Excellent	Excellent	Good	Poor
Inventory control	Excellent	Excellent	Excellent	Poor
Quality	Excellent	Excellent	Good	Fair
Market share	Excellent	Excellent	Good	Fair
Machine utilization	Excellent	Excellent	Good	Poor

Multi-Criteria Decision Making

- Questions to be asked:
  - Basic question: which alternative to select?
  - Other questions:
    - How are the criteria chosen?
    - Who makes the judgments regarding non-monetary criteria such as “quality”?
    - What response is required – a partitioning of alternatives or a rank ordering of alternatives?

Choice of Criteria

- Each criterion distinguishes at least two alternatives (in no case should identical values for an attribute apply to all alternatives).
- Each criterion captures a unique dimension or facet of the decision problem (criteria are independent and non-redundant).
- All criteria, in a collective sense, are assumed to be sufficient for the purpose of selecting the best alternative.
- Differences in values assigned to each criterion are presumed to be meaningful in distinguishing among feasible alternatives.

Choice of Criteria

- New criterion:
  - Cost of operating and maintaining the system.
- Divide "flexibility" in:
  - Ability to interface with computer-aided manufacturing equipment.
  - Capability to create and analyze solid-geometry representations of engineering design concepts.
- Aggregate criteria that have the same values across the four alternatives:
  - "Quality" and "market share" in "achievement of greater market share through quality improvements."

Selection of a Measurement Scale

- Develop metrics that permit various states of each attribute to be represented.

- Example:
  - Cost: $
  - Flexibility: “poor,” “fair,” “good,” “very good,” “excellent.”

Dimensionality of the Decision Problem

- Single-dimensioned analysis.
  - Dimension corresponds to the number of metrics used to represent the attributes that discriminate among alternatives.
  - Collapse all information to a single dimension.
  - Also called compensatory models.
    - Changes in values of a particular attribute can be traded off against opposing changes in another attribute.
- Full-dimensioned analysis.
  - Retain the individuality of the attributes.
  - If *N* attributes chosen to characterize the alternatives under consideration, the predicted values for all *N* attributes are considered in the choice.

Alternative – Attributes Score Card

- Score card.
  - Matrix of alternatives versus criteria, together with numbers or other symbols to represent the outcome expected for each alternative with respect to each attribute.
- Example:

Criterion	Alternative			
	P-1	P-2	P-3	P-4
A. Technical goals	Average	**Above average**	Average	**Below average**
B. Net present worth	$300M	$350M	**$400M**	**$290M**
C. Serviceability	**< 1 hr**	1-1.5 hr	1-1.5 hr	**1.5-2 hr**
D. Management/eng effort	**500 hr**	700 hr	**1,100 hr**	700 hr
E. Lack of riskiness	**Above average**	Average	**Below average**	Average

Best alternative for criterion **Worst alternative for criterion**

Alternative – Attributes Score Card

- Alternative P-3 and P-4 are not as desirable as alternatives P-1 and P-2.
- Can make a final decision based on this method.
  - Weights assigned to the attributes are also important.

Decision Involving Multiple Criteria

- Decisions must be made based on multiple criteria that jointly influence the relative desirability of alternatives under consideration.
- Select criteria to be considered.
  - Criteria need be independent of each other.
  - Number of criteria selected should be kept at a reasonable minimum.
- Differences are the basis of decision.
  - Decisions made only on the basis of their true difference across all criteria.
  - Criteria that are identical are canceled out.

Direct Ranking Methods

- Direct method.
  - Present the important criteria to the decision-maker for ranking.
  - Present alternatives to the decision-maker for ranking.
- Paired comparison method.
  - Consider them two at a time.
  - When there are N alternatives to be ranked, need to compare $N(N-1)/2$ pairs.

Paired Comparison Method

- Example: five criteria to be rank ordered.
  - (1) better, (2) cheaper, (3) faster, (4) repairable, (5) disposable
  - 1<2, 1>3, 1>4, 1=5, 2>3, 2>4, 2>5, 3>4, 3=5, 4>5
  - Transitivity is assumed.
  - Potential lack of consistency.

Criterion	**1**	**2**	**3**	**4**	**5**	**Times preferred**
1	-		P	P	=	2+
2	P	-	P	P	P	4
3			-	P	=	1+
4				-	P	1
5	=		=		-	++

  - Ranked order: 2 > 1 > 3 > 4 = 5

Systematic Elimination Methods

- Values and/or outcomes can be specified for all criteria and all alternatives and should be measurable or at least rank orderable.
- Limitations:
  - Weights that may be applicable to the criteria or attributes are not considered.
  - Possible trade-offs among the criteria across alternatives are not considered.

Systematic Elimination Methods

- Example: four alternatives and four criteria.
  - Ideal and minimum standard specified.

Criterion	Alternatives				Ideal	Minimum Standard
	A	B	C	D		
1 – Better	40	35	50	30	50	30
2 – Cheaper	90	80	75	60	100	70
3 – Faster	6	5	8	6	10	7
4 - Gone	G	P	VG	E	E	F

Systematic Elimination Methods

- Compare alternatives against each other.
  - Check for dominance by making mutual comparisons across alternatives.
  - Just an elimination technique.

Criterion	Alternatives				Ideal	Minimum Standard
	A	B	C	D		
1 – Better	40	35	50	30	50	30
2 – Cheaper	90	80	75	60	100	70
3 – Faster	6	5	8	6	10	7
4 - Gone	G	P	VG	E	E	F

A dominates *B*, so *B* can be eliminated from further consideration.

Systematic Elimination Methods

- Compare alternatives against a standard.
  - Rule 1: alternative may be retained only if it meets the standard for at least one criterion.
  - Rule 2: alternative may be retained only if it meets the standard for all criteria.

Criterion	Alternatives				Ideal	Minimum Standard
	A	B	C	D		
1 – Better	40	35	50	30	50	30
2 – Cheaper	90	80	75	60	100	70
3 – Faster	6	5	8	6	10	7
4 - Gone	G	P	VG	E	E	F

Rule 1: *A*, *B*, *C*, and *D* meet the standard for at least one criterion.

Rule 2: *C* meets the standard for all criteria.

Systematic Elimination Methods

- Compare criteria across alternatives.
  - Rank relevant criteria based on importance.

Ranking of criteria: 2 > 1 > 3 > 4

Systematic Elimination Methods

- First approach:
  - For the most important criterion, choose the best alternative (if any available).
  - Ties between two or more alternatives are broken by using the second most important criterion.
  - Continue until a single alternative survives or until all criteria have been considered.

Criterion	Alternatives				Minimum Standard
	A	B	C	D	
1 – Better	40	35	50	30	30
2 – Cheaper	90	80	75	60	70
3 – Faster	6	5	8	6	7
4 - Gone	G	P	VG	E	F

Most important criterion: 2, for which, best alternative: *A*.

A does not satisfy criterion 3.

Systematic Elimination Methods

- Second approach:
  - Examine one criterion at a time, making comparisons among the alternatives.
  - Continue until all alternatives except one have been eliminated, or until all criteria have been examined.

Criterion	Alternatives				Minimum Standard
	A	B	C	D	
1 – Better	40	35	50	30	30
2 – Cheaper	90	80	75	60	70
3 – Faster	6	5	8	6	7
4 - Gone	G	P	VG	E	F

Criterion	Eliminate	Remaining
1	None	A, B, C, D
2	D	A, B, C
3	A, B, D	C
4	None	C

Select alternative *C*.

Weighting Methods

- Can extend the ranking and elimination methods.
- Provide capability to model priority for criteria.
- If an additive weighting method is used, adding the results together for different criteria permits strength on one criterion to compensate for weakness on another.

Weighting Methods

- Weighting of criteria or attributes.
  - Regardless of whether a direct assignment of weights is made or whether a subjective assignment of importance rating is used, the resulting weights for criteria apply to all alternatives.
  - Resulting weights should sum to 1.00 or 100% for each criterion.

Weighting Methods

- Direct assignment of weights.
  - Assume Better is weighted 35%, Cheaper is weighted 50%, and Faster is weighted 15%.
  - If one of the weights is changed, the others must be changed too (to sum up to 100%).
  - Similarly, 100 points ca be allocated.
- Subjective assignment of importance.
  - Assigned to all alternatives based on a 10-point scale.

Criterion	Importance Rating	Additive Weight
Better	7	7/20 = 0.35
Cheaper	9	9/20 = 0.45
Faster	4	4/20 = 0.20

Weighting Methods

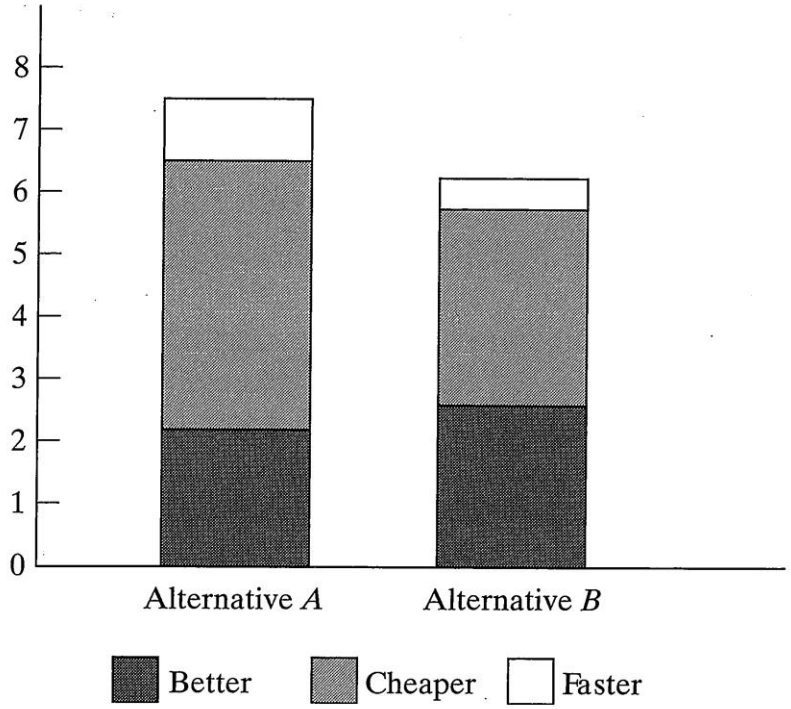
- Tabular additive method.
  - Identifies criteria, weights (*W*), the performance rating (*R*) of each alternative over each criterion, the product of weight and rating (*W* x *R*), and aggregate total of *W* x *R*.

Criterion	**Weight (W)**	**Alternative A**		**Alternative B**	
		Rating (R)	W x R	Rating (R)	W x R
Better	0.35	6	2.10	7	2.45
Cheaper	0.45	10	4.50	6	2.70
Faster	0.20	5	1.00	3	0.60
	∑ 1.00		∑ 7.60		∑ 5.75

Alternative *A* has a higher (*W* x *R*) value.

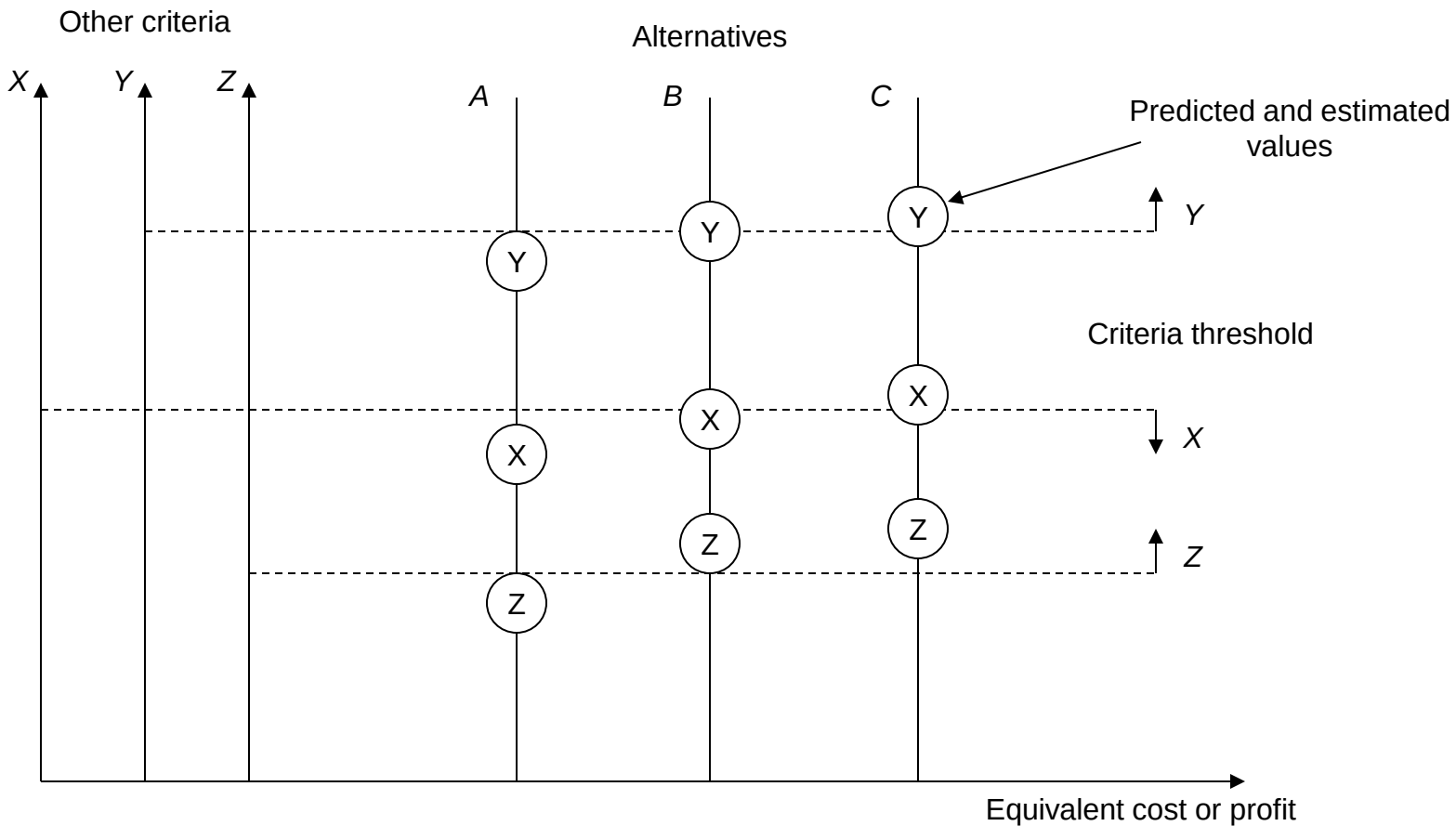
Weighting Methods

- Graphical additive method.
  - Stacked bar chart, with the total height of each bar corresponding to the total score for each alternative.
  - Each bar made up of components that are the product of criterion weights and performance ratings.



Decision Evaluation Display

- Based on differences between alternatives and degree of compliance with multiple criteria.
  - Appropriately exhibited are all that most decision makers need or desire.
  - Simultaneously display the elements pertaining to one or more mutually exclusive alternatives in the face of economic and non-economic criteria.
- Includes:
  - Alternatives (*A*, *B*, *C*).
  - Equivalent cost or profit.
  - Other criteria (*X*, *Y*, *Z*).
  - Other criteria thresholds.
  - Predicted and/or estimated values.

Decision Evaluation Display

- Select the mutually exclusive alternative with the lowest equivalent cost (or maximum equivalent profit) that adequately satisfies the other criteria.